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Enhanced ion transport in PVDF-HFP gel polymer electrolyte containing PDA@BN for lithium ion batteries

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ABSTRACT: Boron nitride nanosheet (BN) modified by polymerized dopamine (PDA@BN) was incorporated into PVDF-HFP gel polymer electrolyte (GPE) by solution casting method. PDA@BN can improve the ionic conductivity ($2.26 \times 10^{-3} \text{ S cm}^{-1}$ at 30 °C), the electrochemical stability of GPE (electrochemical window is 5.0V) and effectively inhibit the growth of lithium dendrites because of barrier effect of two-dimensional BN. The cell prepared with 3P5E2L+6%PDA@BN GPE presents excellent cycle performance. The $\text{LiFePO}_4/\text{3P5E2L+6\%PDA@BN}/\text{Li}$ cells exhibited more than 99% columbic efficiency and the initial discharge capacity reached the maximum of 162.9 mAh/g, capacity retention was about 85% after 80 cycles at 0.1 C. The excellent performance of this GPE implies that it can be applied to high-safety and long-cycle life lithium metal batteries.

Keywords: Boron nitride, Electrolyte, Ion transport, Polymeric composites, Energy storage and conversion

1. Introduction

With the development of technology, there is an increasing demand for high energy density, large capacity electrical energy storage devices [1-3]. Lithium metal is considered to be the most powerful anode candidate for high energy lithium metal battery (LMB) [4-6]. However, uncontrolled lithium dendrites formed during charge and discharge and poor cycle efficiency make LMB uncommercial [7-9]. Gel polymer

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electrolytes (GPEs) is considered to be the most attractive alternative to traditional carbonate-based liquid electrolytes because GPEs can effectively inhibit leaking, flammability, and the growth of lithium dendrites of LMB [10,11]. PVDF-HFP is widely used in GPEs due to high dielectric constant, mechanical properties and electrochemical stability by solution casting method [12,13]. Boron nitride (BN) in GPEs can effectively inhibit the growth of lithium dendrites because BN provides protection for the lithium metal anode during battery cycling [14].

In this work, a series of GPEs containing PVDF-HFP, LiTFSI, [EMIM][TFSI], PDA@BN were prepared by simple solution casting method. PDA@BN in GPEs forms more ion channels, improves electrochemical stability, and inhibit the growth of lithium dendrites. The effects of PDA@BN on the electrochemical performance, including ionic conductivity, electrochemical stability, interfacial behavior and battery performance for these electrolytes were investigated and compared in detail.

2. Experimental

2.1. Materials

PVDF-HFP ($M_w=400,000$) was purchased from Sigma-Aldrich. LiTFSI was purchased from Shanghai Macklin Biochemical Co, Ltd. [EMIM][TFSI] was obtained from Lanzhou Greenchem ILs Co, Ltd. Boron nitride nanosheet (BN) were provided by Nanjing XFNANO Material Technology Co., Ltd., China. The preparation and characterization of PDA@BN are showed in Fig. S1. All other reagents and solvents were obtained from reliable commercial sources and used as received.

2.2. Preparation of GPEs

Different weight ratios (shown in Table S1) of PVDF-HFP, [EMIM][TFSI] and LiTFSI were mixed and dissolved in DMF by magnetic stirring at 80 °C for 4 h. Different amounts of PDA@BN were dispersed in DMF by ultrasonication and then added in the above solution and again stirred for 2 h at 80 °C. The obtained dispersion liquid was cast on a Teflon mold and the membranes were dried at 110 °C for 12 h.

2.3. Characterization

The ionic conductivity (σ) and interfacial compatibility of GPEs were measured by electrochemical impedance spectrum (EIS) analysis, using an electrochemical workstation (CHI660B). The battery cycle test was evaluated by assembling the GPEs between LiFePO_4 and lithium electrodes form a $\text{LiFePO}_4/\text{GPE}/\text{Li}$ cell with the cut off voltage of (4.2-2.7 V vs. Li/Li^+) on a Land battery cycler.

3. Results and Discussion

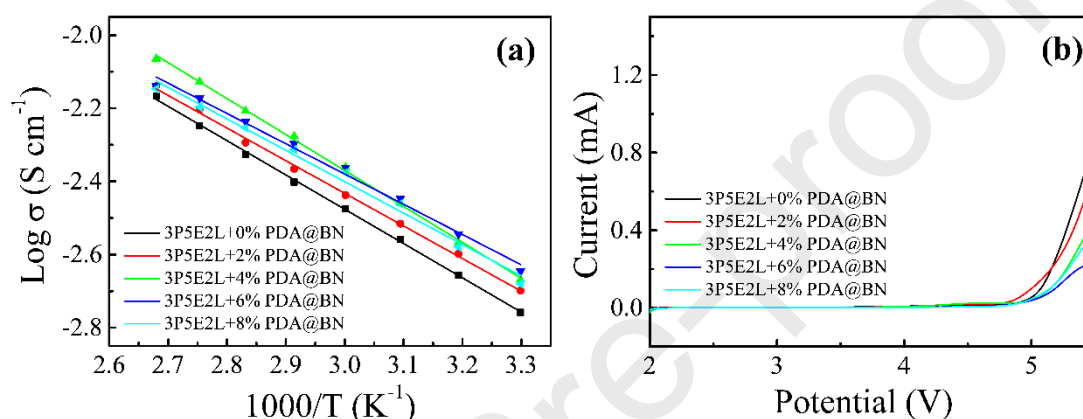


Fig. 1 (a) Temperature dependence of ionic conductivity of all GPEs. (b) The linear sweep voltammetry curve with a ss/GPEs/Li cell at room temperature.

Fig. 1a shows the temperature dependence on the ionic conductivity (σ) for different GPEs. The 3P5E2L+6%PDA@BN shows the highest conductivity at room temperature, reaching $2.26 \times 10^{-3} \text{ S cm}^{-1}$. Compared with 3P5E2L+0%PDA@BN, the σ ($1.65 \times 10^{-3} \text{ S cm}^{-1}$) of 3P5E2L+6%PDA@BN is increased by 140%. The nano-structured and surface-functional PDA@BN shows greater dispersibility and compatibility with the PVDF-HFP matrix, forming more ion channels and enhancing more ion transmission [2,15]. The large conductivity enhancement is ascribed to fast ion-conducting pathway without crossing junctions on the surfaces of the BN. However, the σ of 3P5E2L+8%PDA@BN decreases slightly due to the excess barrier effect of BN with two-dimensional structure on the conductive channels [16]. The calculated activation energy (E_a) value of 3P5E2L+0%PDA@BN is 17.9 kJ/mol. When 6% PDA@BN was added, the E_a decreases to 15.8 kJ/mol due to the increase

of ion channels. But the E_a of 3P5E2L+8%PDA@BN increases to 16.4 kJ/mol because of the excess barrier effect of BN. The lithium ion transference number of 3P5E2L+6%PDA@BN is bigger than that of 3P5E2L+0%PDA@BN because PDA@BN improves ion transmission and interface compatibility (Fig. S2).

The Fig. 1b shows the linear sweep voltammetry (LSV) of an asymmetric battery of Li/3P5E2L+x%PDA@BN/SS. With the increase of PDA@BN in GPEs, the electrochemical stability window of GPEs firstly decreases and then increases. This indicates that 2% PDA@BN does not form barrier effect of two-dimensional BN, and the oxidation of polydopamine leads to a reduction in LSV. When the content of PDA@BN is beyond 2% PDA@BN, barrier effect of BN can form, PDA@BN can effectively suppress the oxidation degree of anions at high potential and improve electrochemical stability [14]. The electrochemical stability window of PDA@BN modified GPE is more than 5.0 V, which lays a foundation for the application of GPE in high voltage battery.

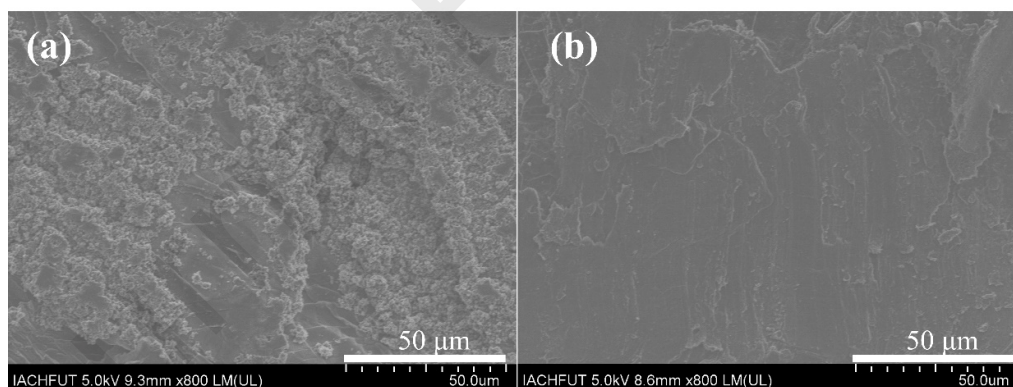


Fig. 2 The morphology of Li anode in Li/GPEs/LiFePO₄ cell with different electrolyte after 80 cycles. (a) 3P5E2L+0%PDA@BN GPE; (b) 3P5E2L+6%PDA@BN GPE.

The compatibility of GPE with lithium metal was analyzed by evaluating the change in EIS of a symmetric Li/GPE/Li cell for 30 days (Fig. S3). It is clear that the 3P5E2L+6%PDA@BN GPE exhibits a more gentle change in interface impedance compared to other electrolytes. The 6% PDA@BN improves the ionic conductivity and facilitates the transmission of lithium ions. The further reactions between GPE

and lithium anode is suppressing. The impedance before and after the battery cycling also shows the same trend (Fig. S4). The morphology of lithium anode with different GPE after cycling were compared by SEM showed in Fig. 2. After 80 cycles of charge and discharge cycles, the surface of the lithium anode of 3P5E2L+0%PDA@BN GPE was severely damaged, and there was obvious lithium dendrite growth. However, the surface of the lithium anode of 3P5E2L+6%PDA@BN GPE was slightly damaged due to the inevitable release and insertion of lithium ions, and no obvious lithium dendrites appeared. The high ionic conductivity and good interfacial compatibility improved the speed of lithium ion transport. Barrier effect of two-dimensional BN can hinder the growth of lithium dendrites. 3P5E2L+0%PDA@BN cell short-circuited after 100h, but 3P5E2L+6%PDA@BN cell worked stably within 300h (Fig. S5). It is indicated that 3P5E2L+6%PDA@BN GPE has a good inhibitory effect on lithium dendrites growth.

Fig. 3 shows the charge-discharge curves of the Li/3P5E2L+0%PDA@BN/LiFePO₄ cell and Li/3P5E2L+6%PDA@BN/LiFePO₄ cell cycled at 30 °C in the voltage range 2.7 V to 4.2 V at a rate of 0.1 C. It clearly shows a voltage plateau for charge (3.39 V) and discharge (3.46 V) for two groups of batteries. The Li/3P5E2L+0%PDA@BN/LiFePO₄ cell showed worse cycle stability than the Li/3P5E2L+6%PDA@BN/LiFePO₄ cell, as shown in Fig. 3c. The Li/3P5E2L+0%PDA@BN/LiFePO₄ cell showed an initial discharge capacity of 144.3 mAh/g and a capacity retention rate of 74% after 80 charge and discharge cycles, while its coulomb efficiency was only 96%. In contrast, the introduction of 6% PDA@BN in the GPE can significantly improve battery performance. After 80 charge and discharge cycles, the 3P5E3L+6%PDA@BN cell had an initial discharge capacity of 162.9 mAh/g and a capacity retention rate of 85%, and the Coulomb efficiency remained above 99% throughout the cycle. The high discharge capacity and capacity retention were due to the higher ionic conductivity and good interfacial compatibility [18].

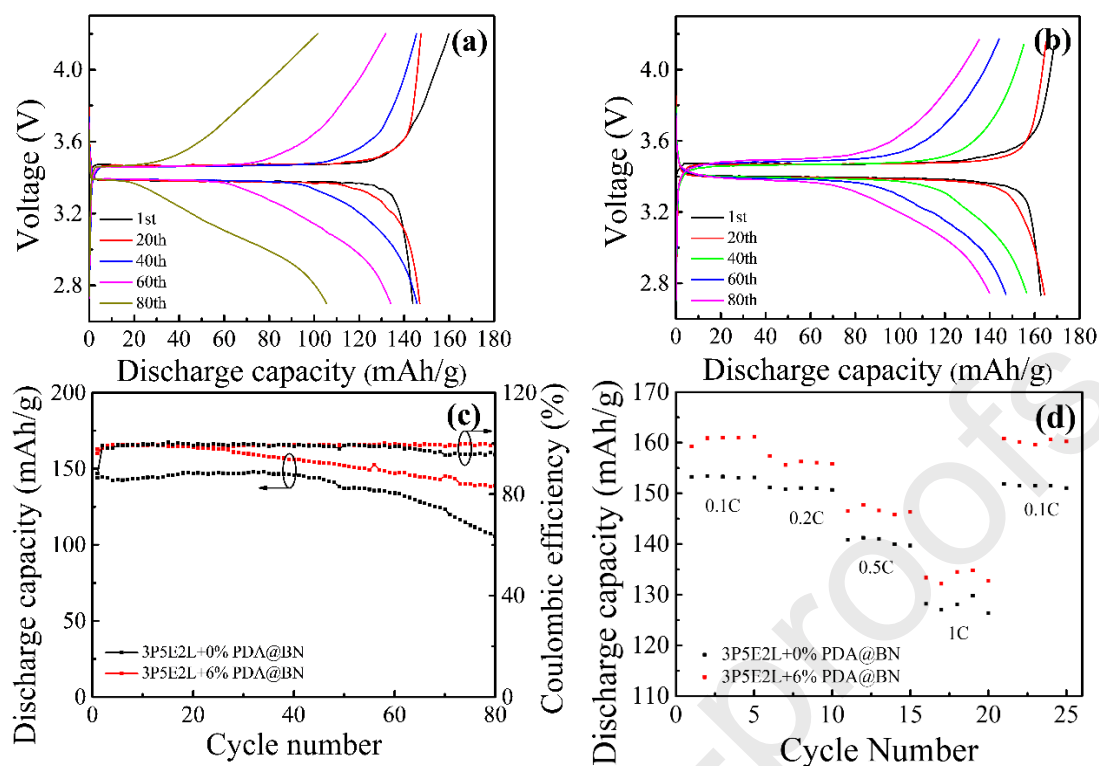


Fig. 3 Charge/discharge curves at 30 °C with 0.1C current rate, (a) 3P5E2L+0% PDA@BN, (b) 3P5E2L+6%PDA@BN; (c) Discharge capacity and coulombic efficiency; (d) Discharge capacity at various current rates.

The rate cycling performance of 3P5E2L+6%PDA@BN cell at different current densities shows in Fig. 3d. It can be clearly seen that the specific discharge capacities of all electrolytes gradually decreased with increasing current density. This was due to the increased interface resistance and polarization effects at higher current densities, limiting the diffusion of Li^+ [19]. The Li/3P5E2L+6%PDA@BN/LiFePO₄ cell exhibited high discharge capacity of approximately 161, 156, 146 mAh/g and 133 mAh/g at high rates of 0.1 C, 0.2 C, 0.5 C and 1 C, respectively. In contrast, the 3P5E2L+0%PDA@BN cell had lower specific discharge capacity at corresponding current density. These results show that 3P5E2L+6% PDA@BN cell presented good cycling performance and excellent rate performance.

4. Conclusion

The 3P5E2L+6%PDA@BN GPE can effectively improve ionic conductivity,

electrochemical stability, and interface contact between the electrolyte and the electrode. The battery **which was** composed of lithium metal anode and LiFePO_4 cathode using 6% PDA@BN enhanced polymer/salt hybrid electrolyte **presented** improved cycling performance (140 mAh/g after 80 cycles at 0.1C), with high coulomb efficiency (over 99%). **The** 6% PDA@BN can effectively inhibit lithium dendrite growth during stripping and plating **because of the barrier effect of two-dimensional BN.**

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Author Statement

Hanyang Chen: Writing original draft.

Pei Xu: Writing - Review & Editing.

Lang Chen: Investigation.

Xueliang Li: Writing - Review & Editing.

Yunsheng Ding: Supervision.

Conflict of interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Declaration of Interest Statement

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in the manuscript entitled.

Highlights

- Barrier effect of two-dimensional BN can inhibit the growth of lithium dendrites.
- PDA@BN can promote uniform dispersion of BN and stabilize lithium metal.
- The 3P5E2L+6%PDA@BN GPE can effectively enhance ion transmission.